

Optimization with ordering

Ordered Operators and Decision Making (II)

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Doc-Course Optimization over Nonlinear Model Spaces
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Incorporating Ordering into Optimization Problems

Scenario Analysis in Linear Programming

Weighted Set Covering

Traveling SalesMan Problem

Fitting Linear Models

LP Relaxations of Ordered Problems

Incorporating Ordering into Optimization Problems

Embedding Ordered Measures in Optimization

The main advantage of optimization-based representations is that ordered measures can be used when $\mathbf{x}$ is itself decision-dependent.

$$\mathbf{x} = \mathbf{x}(y) \quad \Longrightarrow \quad \min_{y \in \mathcal{Y}} \Phi(\mathbf{x}(y)),$$

where Φ may be one of

$$\ell(\mathbf{x}, \lambda), \quad \mathcal{Q}(\mathbf{x}, M), \quad \mathfrak{b}(\mathbf{x}, \rho).$$

- Sorting evaluates a measure for a fixed vector.
- Optimization models search over many possible vectors $\mathbf{x}(y)$.
- This allows ordered aggregation to guide decision making.

A General Modeling Template

Let $\mathcal{Y}$ be the feasible set of a decision problem and let

$$\mathbf{x}(y) = (x_1(y), \dots, x_n(y)).$$

We can solve

$$\begin{array}{ll} \min & \Phi(\mathbf{x}(y)) \\ \text{s.t.} & y \in \mathcal{Y}, \end{array}$$

with different choices of Φ :

- $\Phi = \ell$: ordered location or ordered risk criterion.
- $\Phi = \mathcal{Q}$: ordered dispersion or interaction criterion.
- $\Phi = \mathfrak{b}$: deviation around an ordered location.

Modeling issue

When $\mathbf{x}$ depends on decision variables, the resulting model may include bilinear or nonlinear terms.

Handling Nonlinearities

The embedded formulations may become nonlinear.

Binary products

If products involve binary variables, use exact McCormick linearizations.

- Some formulations remain linear.
- Some become convex quadratic or conic.
- Others are genuinely nonconvex.

The ordered measure is no longer only an evaluation tool; it becomes part of the optimization model.

Continuous nonlinearities

If the model is nonconvex, use global optimization tools such as spatial branch-and-bound.

Scenario Analysis in Linear Programming

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Application 1: Scenario Analysis in Linear Programming

Consider a polyhedron

$$P = \{z \in \mathbb{R}_+^d : Az \leq b\}$$

and a matrix of scenario costs

$$C \in \mathbb{R}^{n \times d}.$$

For each feasible solution $z \in P$, define

$$\mathbf{x} = Cz \in \mathbb{R}_+^n.$$

Application 1: Scenario Analysis in Linear Programming

$$\min_{z \in P} \ell(Cz, \lambda), \quad \min_{z \in P} Q(Cz, M), \quad \min_{z \in P} b(Cz, \rho).$$

Interpretation

Different ordered criteria select different robust, balanced, or risk-sensitive solutions.

Scenario Analysis: Toy Polyhedron

We consider a planar polyhedron with vertices

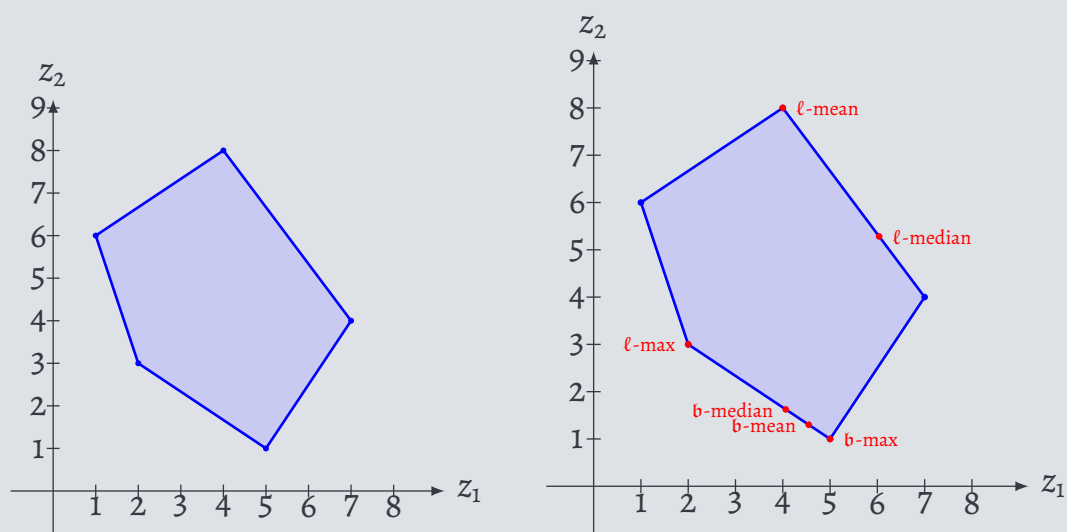
$$V = \{(2, 3), (5, 1), (7, 4), (4, 8), (1, 6)\}$$

and cost matrix

$$C = \begin{pmatrix} 3 & 4 \\ 2 & 0 \\ -3 & -2 \\ -2 & -6 \\ 4 & -10 \end{pmatrix}.$$

- Each $z \in P$ induces five scenario values.
- Ordered measures aggregate the vector Cz .
- Different measures identify different points of the same feasible region.

Scenario Analysis: Different Ordered Criteria



Weighted Set Covering

Application 2: Weighted Set Covering

Let $M = \{1, \dots, m\}$ be a set of items and let

$$\mathcal{S} = \{S_1, \dots, S_n\}, \quad S_i \subseteq M.$$

Each item $j \in M$ has a weight $w_j \geq 0$.

$$y_i = \begin{cases} 1, & \text{if subset } S_i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

The item-wise induced cost is

$$x_j(y) = w_j \sum_{i: j \in S_i} y_i, \quad j \in M.$$

Weighted Set Covering Model

The classical weighted covering model is

$$\begin{aligned} \min \quad & \sum_{j \in M} w_j \sum_{i: j \in S_i} y_i \\ \text{s.t.} \quad & \sum_{i: j \in S_i} y_i \geq 1, & j \in M, \\ & y_i \in \{0, 1\}, & i = 1, \dots, n. \end{aligned}$$

Equivalently,

$$\min \sum_{i=1}^n \left(\sum_{j \in S_i} w_j \right) y_i.$$

- The standard objective aggregates item costs by the sum.
- Ordered measures allow equity, robustness, and tail-sensitive criteria.

Weighted Set Covering with Ordered Objectives

For the induced vector

$$x_j(y) = w_j \sum_{i: j \in S_i} y_i,$$

we solve

Linear

$$\min_y \ell(\mathbf{x}(y), \lambda)$$

Nested

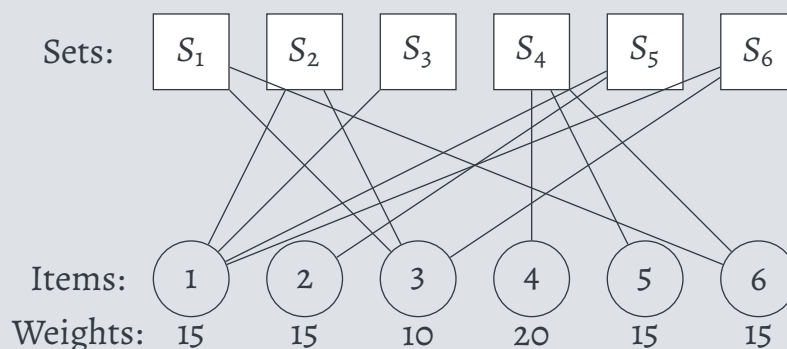
$$\min_y \mathfrak{b}(\mathbf{x}(y), \rho)$$

Quadratic

$$\min_y Q(\mathbf{x}(y), M)$$

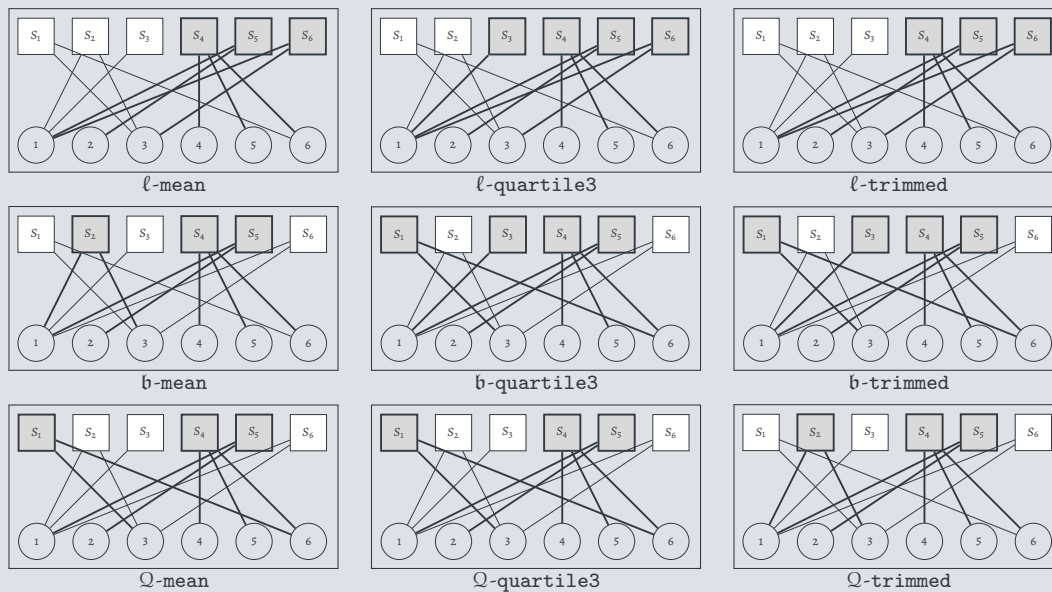
- ℓ : mean, trimmed mean, third quartile.
- $\mathfrak{b}$: absolute deviations around these ordered locations.
- Q : variance, squared deviations, trimmed variance.

Weighted Set Covering: Instance



- Subsets are shown at the top.
- Items are shown at the bottom with their weights.
- Edges indicate item membership in subsets.

Weighted Set Covering: Ordered Solutions



Weighted Set Covering: Interpretation

Even in a small illustrative instance, different ordered objectives select different subsets.

- Mean-type criteria favor low total weighted coverage.
- Quartile-type criteria control the upper part of the cost distribution.
- Trimmed criteria ignore some extreme values.
- Deviation-based criteria promote balanced item-wise costs.
- Quadratic criteria penalize dispersion more strongly.

Traveling SalesMan Problem

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Application 3: Traveling Salesman Problem

Let $G = (V, A)$ be a directed graph with arc weights $w_a \geq 0$.

Introduce binary variables

$$z_a = \begin{cases} 1, & \text{if arc } a \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Since each node has exactly one outgoing arc in the tour, define

$$x_v(z) = \sum_{a \in \delta^+(v)} w_a z_a, \quad v \in V.$$

Classical TSP

Minimizing total tour length is equivalent to minimizing $\sum_{v \in V} x_v(z)$.

Ordered Objectives for the TSP

Instead of minimizing total length, we aggregate the vector of outgoing arc lengths:

$$\mathbf{x}(z) = (x_v(z))_{v \in V}. \text{ (see (Ljubić et al., 2026))}$$

Robustness

$$\min \max_v x_v(z)$$

Balance

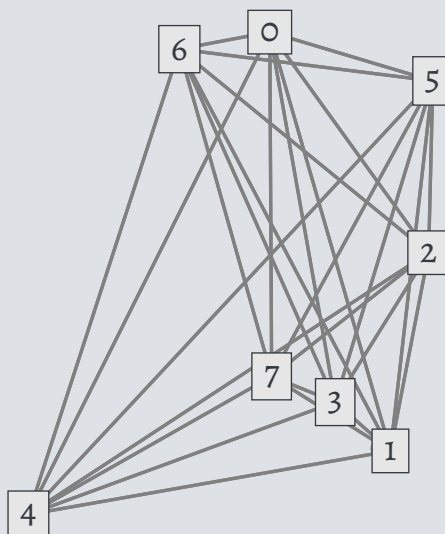
$$\min \mathfrak{b}(\mathbf{x}(z), \rho)$$

Dispersion

$$\min \mathcal{Q}(\mathbf{x}(z), M)$$

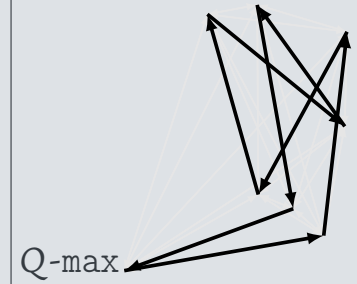
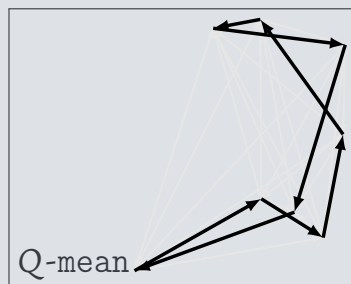
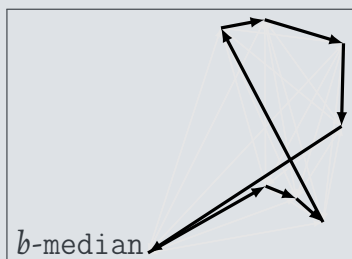
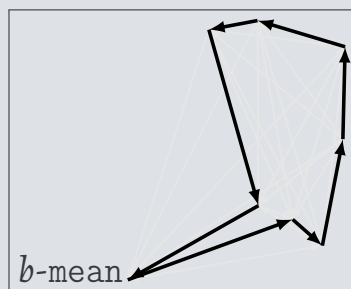
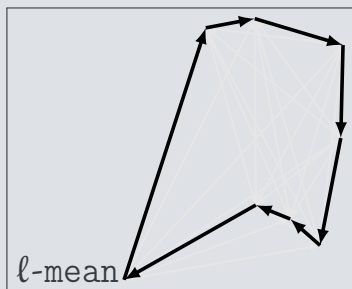
- The classical solution minimizes average length.
- Ordered objectives may avoid very long individual legs.
- Deviation criteria may produce tours with more homogeneous arc lengths.

TSP Example



- The graph is complete.
- Arc costs are Euclidean distances between points.
- We compare ℓ , $\mathfrak{b}$, and $\mathcal{Q}$ ordered objectives.

TSP: Solutions under Ordered Measures



TSP: Interpretation

- The classical TSP corresponds to ℓ -mean.
- Minimizing ℓ -max yields a robust tour by controlling the largest outgoing leg.
- Deviation-based objectives avoid tours with highly uneven arc lengths.
- Quadratic objectives penalize large deviations more strongly than absolute deviations.
- Some ordered solutions may have larger total length but better balance across legs.

Fitting Linear Models

⊂ {V. Blanco, A. Japón, J. Puerto, D. Ponce, R. Salmerón}

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Fitting Linear Models

Fitting Hyperplane

Given $\{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$, find $\beta_0, \beta_1, \dots, \beta_d \in \mathbb{R}$ such that:

$$\beta_0 + \beta_1 x_{i1} + \dots + \beta_d x_{id} \approx 0.$$

Two main ingredients for the estimation of β :

- **Residuals:** For a single observation $x_i \in \mathbb{R}^d$, individual measures of error ($\varepsilon_{x_i}(\beta)$).
- **Aggregation:** Given $\varepsilon_{x_i}, \forall i$, an overall measure of the global error: $\Phi(\varepsilon_{x_1}, \dots, \varepsilon_{x_n})$

Estimation of a Fitting Hyperplane

$$\min_{\beta \in \mathbb{R}^{d+1}} \Phi(\varepsilon_{x_1}(\beta), \dots, \varepsilon_{x_n}(\beta))$$

Fitting Hyperplanes: Residuals

Residuals

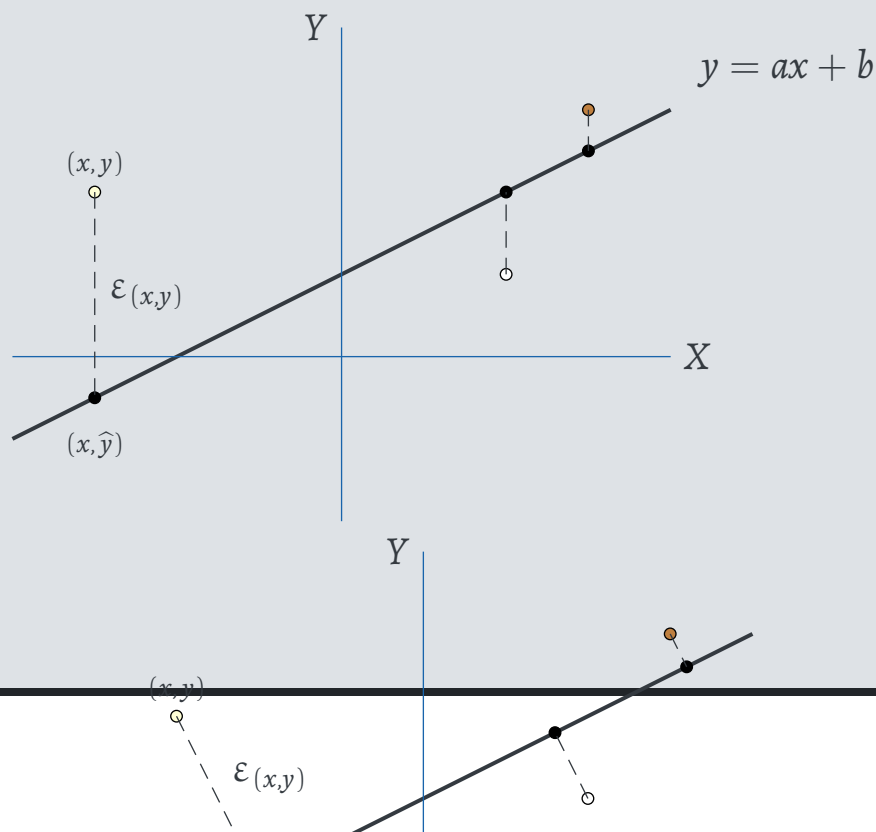
The residual represents how “far” is the point (observation) $x \in \mathbb{R}^{d+1}$ with respect to the hyperplane $\mathcal{H}(\beta)$.

We model the **point-to-hyperplane deviation** by a residual mapping:

$$\varepsilon_x : \mathbb{R}^{d+1} \rightarrow \mathbb{R}_+, \quad \varepsilon_x(\beta) = D(x, \mathcal{H}(\beta))$$

being D a distance measure in $\mathbb{R}^d$.

Fitting Hyperplanes: Residuals



Fitting Hyperplanes: Residuals

Mangasarian, 99

For a given point $z^t = (1, z_1, \dots, z_d)$ and the hyperplane $\mathcal{H}(\beta)$:

$$\varepsilon_z = \min_{y \in \mathcal{H}(\beta)} \|z - y\| = \frac{|\beta^t z|}{\|\beta_{-0}\|^*}.$$

where $\|y\|^* = \max_{z \in \mathbb{R}^d: \|z\| \leq 1} z^t y$, $k(\beta) = \arg \max_{\|x\|=1} \beta_{-0}^t x$

Response: $\hat{z}_d = -\frac{\beta^t z}{\|\beta_{-0}\|^*} k(\beta)_d$.

Marginal variation: $\frac{\partial \hat{z}_d}{\partial z_j} = -\frac{\beta_j}{\|\beta_{-0}\|^*} k(\beta)_d$,

Fitting Hyperplanes: Residuals

1. If D is the ℓ_1 - distance,

$$\hat{z}_k = \begin{cases} z_k & \text{if } |\beta_k| \neq \max\{|\beta_j| : j = 1, \dots, d\}, \\ z_k - \frac{\beta^t z}{\|\beta_{-0}\|_1} v_k, & \text{if } \beta_k = \max\{|\beta_j| : j = 1, \dots, d\}, \\ z_k + \frac{\beta^t z}{\|\beta_{-0}\|_1} v_k, & \text{if } \beta_k = -\max\{|\beta_j| : j = 1, \dots, d\}, \end{cases}$$

for $k = 1, \dots, d$, and for some $v_1, \dots, v_d \geq 0$ such that $\sum_j v_j = 1$.

2. If D is the ℓ_∞ - distance,

$$\hat{z}_k = \begin{cases} z_k - \frac{\beta^t z}{\|\beta_{-0}\|_1}, & \text{if } \beta_k > 0, \\ z_k + \frac{\beta^t z}{\|\beta_{-0}\|_1}, & \text{if } \beta_k < 0, \end{cases} \quad k = 1, \dots, d.$$

3. If D is the ℓ_τ - distance with $1 < \tau < +\infty$ then

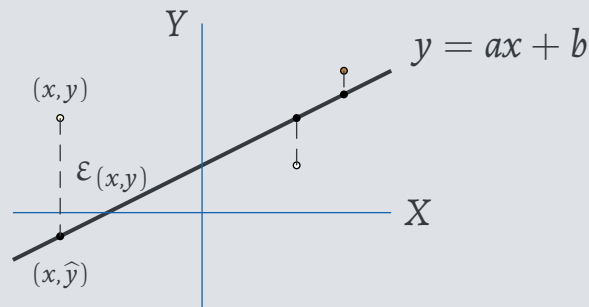
$$\hat{z}_k = z_k - \frac{\beta^t z}{\|\beta_{-0}\|_\tau} k_\tau(\beta)_k, \quad k = 1, \dots, d$$

and

$$k_\tau(\beta)_k = \begin{cases} \frac{(\beta_k)|\beta_k|^{v/\tau}}{(\sum_{j=1}^d |\beta_j|^v)^{1/\tau}} & \text{if } \beta_k \neq 0 \\ 0 & \text{if } \beta_k = 0, \end{cases} \quad k = 1, \dots, d,$$

being v such that $\frac{1}{\tau} + \frac{1}{v} = 1$.

Fitting Hyperplanes: Aggregation



Given residuals $(\varepsilon_1(\beta), \dots, \varepsilon_n(\beta)) \rightarrow$ MO Problem (Carrizosa et. al, 1995).

Given a set of residuals $\varepsilon_1, \dots, \varepsilon_n \geq 0$ aggregated using a function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$, the Fitting Hyperplane Problem consist of finding $\hat{\beta} \in \mathbb{R}^{d+1}$ such that:

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^{d+1}} \Phi(\varepsilon(\beta)),$$

Estimation of a Fitting Hyperplane

$$\min_{\beta \in \mathbb{R}^{d+1}} \Phi(\varepsilon_{x_1}(\beta), \dots, \varepsilon_{x_n}(\beta))$$

$$\min_{\beta} \text{OM}_{\lambda}(D(x_1, \mathcal{H}(\beta)), \dots, D(x_n, \mathcal{H}(\beta)))$$

$$\min \sum_{j=1}^n v_j + \sum_{i=1}^n w_i$$

$$\text{s.t. } \varepsilon_i \geq \frac{|\beta^t x_i|}{\|\beta - o\|^*}, \quad \forall i = 1, \dots, n,$$

1. Least Sum of Squares (Gauss, 1809)

$$\Phi_{LSS}(\varepsilon_1, \dots, \varepsilon_n) = \sum_{i=1}^n \varepsilon_i^2,$$

$\lambda^t = (1, \dots, 1), p = 2$ and ε the vertical distance.

2. Least Absolute Deviation (Edgeworth, 1887)

$$\Phi_{LAD}(\varepsilon_1, \dots, \varepsilon_n) = \sum_{i=1}^n |\varepsilon_i|.$$

$\lambda^t = (1, \dots, 1), p = 1$ and ε the vertical distance.

3. Least Quantile of Squares (Bertsimas & Mazumder, 2014; Hampel, 75):

$$\Phi_{LQS}(\varepsilon_1, \dots, \varepsilon_n) = r - \text{quantile}(\varepsilon_1^2, \dots, \varepsilon_n^2) := \varepsilon_{(r)}^2.$$

$$\lambda = (\overbrace{0, \dots, 0}^{(r-1)}, 1, \overbrace{0, \dots, 0}^{(n-r-1)}), p = 2 \text{ and } \varepsilon \text{ the vertical distance.}$$

4. The Least Trimmed Sum of Squares (Rousseeuw, 84):

$$\Phi_{LTS}(\alpha)(\varepsilon_1, \dots, \varepsilon_n) = \sum_{i=1}^{\lfloor \alpha n \rfloor} \varepsilon_{(i)}^2,$$

$$\lambda = (\overbrace{0, \dots, 0}^{(r-1)}, \overbrace{1, \dots, 1}^{\lfloor \alpha n \rfloor}), p = 2 \text{ and } \varepsilon \text{ the vertical distance.}$$

Multisource Hyperplane Location



On the multisource hyperplanes location problem to fitting set of points
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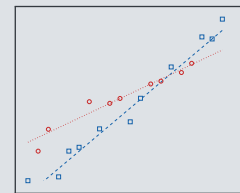
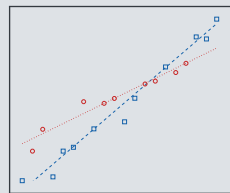
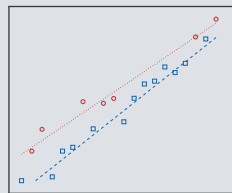
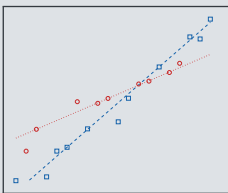
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Mixed integer linear programming
Column generation

ABSTRACT

In this paper we study the problem of locating a given number of hyperplanes minimizing an objective function of the closest distances from a set of points. We propose a general framework for the problem in which norm-based distances between points and hyperplanes are aggregated by means of ordered median function. A compact Mixed Integer Linear (or Non Linear) programming formulation is presented for the problem and also an extended set partitioning formulation with a huge number of variables is derived. We develop a column generation procedure embedded within a branch-and-price algorithm for solving the problem by adequately performing its preprocessing, pricing and branching. We also analyze geometrically the optimal solution of the problem, deriving properties which are exploited to generate initial solutions for the proposed algorithm. Finally, the results of an extensive computational experience are reported. The issue of scalability is also addressed showing theoretical upper bounds on the errors assumed by replacing the original datasets by aggregated versions.

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- p hyperplanes located.
- Assignments to *closest* hyperplane.
- Much more difficult to solve.
- Generalizes ClusterWise/PieceWise linear fittings.
- A Column generation approach.
- Scalability Results



LP Relaxations of Ordered Problems

Víctor Blanco, Diego Laborda, Miguel Martínez-Antón

<https://arxiv.org/abs/2510.09166>

Discrete Ordered Median Problem

Given a finite subset P of a metric space $(\mathcal{X}, d)$, and a weight vector $\lambda \in \mathbb{R}^{|P|}$

The goal of a p -DOMP is to find a subset $Q \subseteq P$ with $|Q| = p$ that minimizes an ordered median

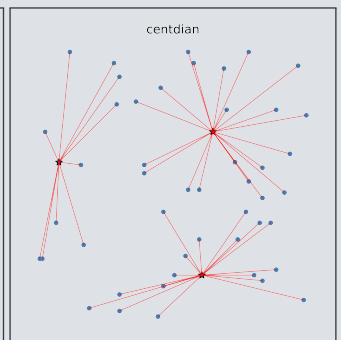
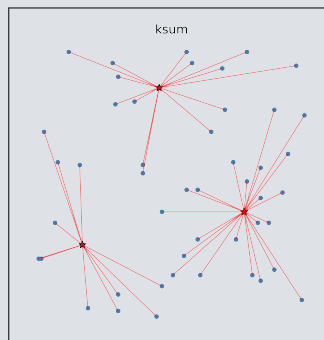
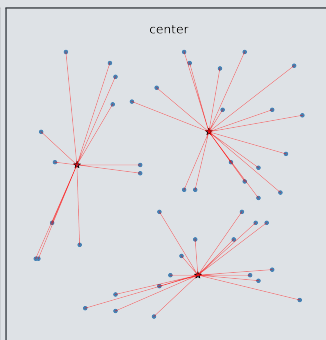
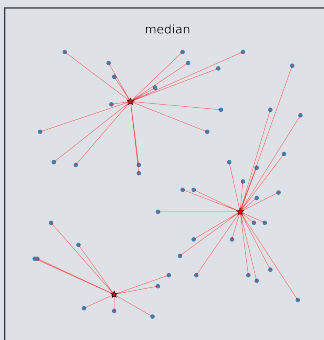
$$\min_{\substack{Q \subseteq P \\ |Q|=p}} \ell_{\lambda}(Q) = \sum_{j=1}^{|P|} \lambda_j \min_{b \in Q} d(\sigma(j), b) \quad (p\text{-DOMP})$$

$\sigma : [|P|] \rightarrow P$ is an ordering such that $\min_{b \in Q} d(\sigma(j), b) \geq \min_{b \in Q} d(\sigma(j+1), b)$ for $j \in [|P| - 1]$

- Facility Location: Nickel and Puerto (2005); Blanco et al. (2014); Blanco et al. (2016); Labbé et al. (2017); Marín et al. (2020); Ljubić et al. (2024), ...
- Voting Ponce et al. (2018), Portfolio Cesarone and Puerto (2024), Net Design Puerto et al. (2016), ML Blanco et al. (2018); Martinez-Merino et al. (2025), Insurance Blanco and Pérez-Sánchez (2018), ...

Ordered Location and Clustering

λ	Problem	Acronym	Description
$(1, \dots, 1)$	p -median	M_p	Sum of all distances
$(1, 0, \dots, 0)$	p -center	C_p	maximum distance
$(1, \dots, 1, 0, \dots, 0)$	p - k -sum	S_p^k	sum of k -largest distances
$(1, \gamma, \dots, \gamma)$	p - γ -centdian	D_p^{γ}	Convex combination of max-sum



MILP formulations (non-increasing λ weights)

Blanco et al. (2014)

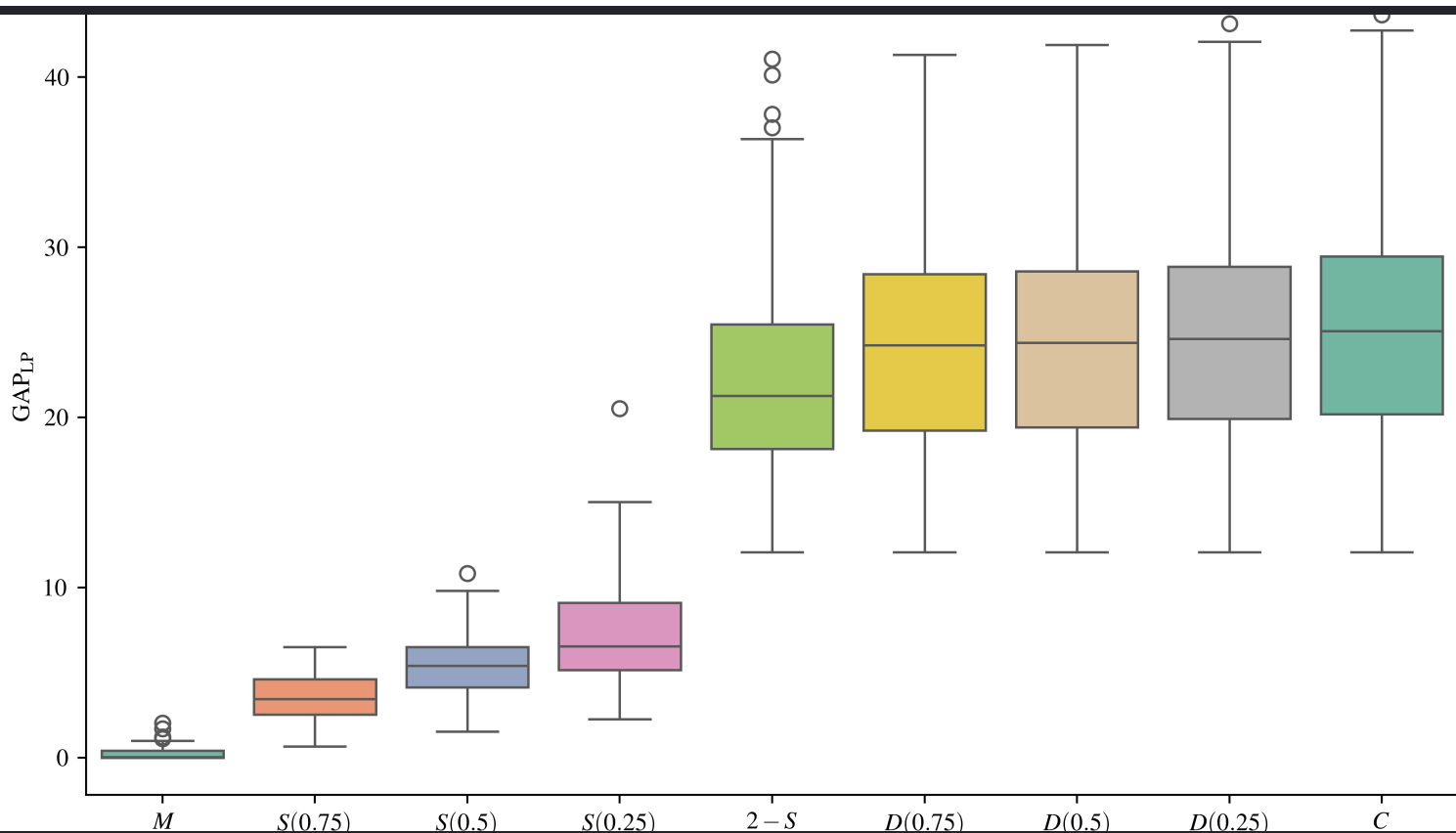
$$\begin{aligned} \min \quad & \sum_{i=1}^m u_i + \sum_{r=1}^m v_r \\ & \sum_{j=1}^m z_{ij} = 1, & \forall i \in [m], \\ & z_{ij} \leq y_j, & \forall i, j \in [m], \\ & \sum_{j=1}^m y_j = p, \\ & u_i + v_r \geq \lambda_r \sum_{j=1}^m z_{ij} d_{ij}, & \forall i, r \in [m], \\ & u_i, v_r \in \mathbb{R}, & \forall i, r \in [m], \\ & y_j, z_{ij} \in \{0, 1\} [0, 1], & \forall i, j \in [m]. \end{aligned}$$

Ogryczak and Tamir (2003)

$$\begin{aligned} \min \quad & \sum_{s=1}^m \Delta_s \left(s t_s + \sum_{i=1}^m q_{is} \right) \\ & \sum_{j=1}^m z_{ij} = 1, & \forall i \in [m], \\ & z_{ij} \leq y_j, & \forall i, j \in [m], \\ & \sum_{j=1}^m y_j = p, \\ & q_{is} \geq \sum_{j=1}^m d_{ij} z_{ij} - t_s, & \forall i, s \in [m], \\ & t_s \in \mathbb{R}, & \forall s \in [m], \\ & q_{is} \in \mathbb{R}, & \forall i, s \in [m], \\ & y_j, z_{ij} \in \{0, 1\} [0, 1], & \forall i, j \in [m]. \end{aligned}$$

Under some conditions... **p-median can be recovered by its LP relaxation** see, e.g., Del Pia and Ma (2023)

$$\Delta_s := (\lambda_s - \lambda_{s+1}) \text{ if } s < m \text{ and } \Delta_m := \lambda_m$$

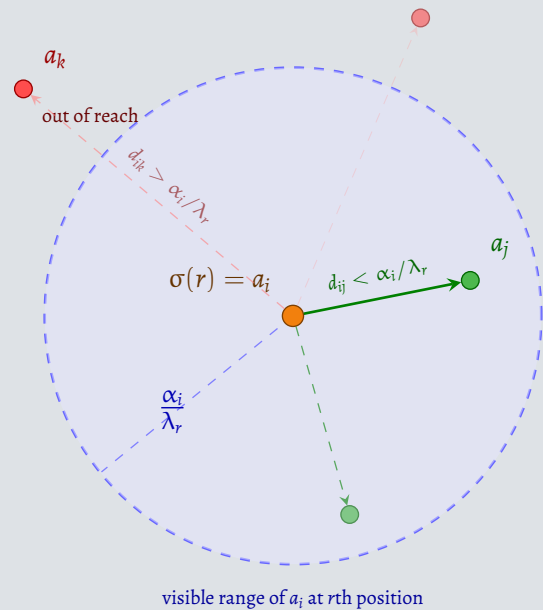


Ordered Contribution Function

Let $\alpha \in \mathbb{R}^m$ and $\sigma \in \mathbb{R}^{m \times m}$ be a permutation matrix. The *ordered contribution function*, $C_\sigma^\alpha : [m] \rightarrow \mathbb{R}_+$, is defined by

$$C_\sigma^\alpha(j) := \sum_{i=1}^m \left(\alpha_i - \sum_{r=1}^m \lambda_r \sigma_{ir} d_{ij} \right)_+$$

- The overall ordered contribution attains its maximum in the centers
- Each point *sees* exactly its own center



Necessary and Sufficient Condition for Exactness

Let $(\bar{y}, \bar{z}, u, v)$ be a feasible solution to DOMP_{IP} . If $(\bar{y}, \bar{z}, u, v)$ is optimal for DOMP_{LP} then:
 $\exists \alpha \in \mathbb{R}^m, \sigma \in \mathbb{R}^{m \times m}$, such that

$$C_\sigma^\alpha(j) = C_\sigma^\alpha(j'), \quad \forall j, j' \in Q; \quad (1)$$

$$C_\sigma^\alpha(i) \leq C_\sigma^\alpha(j), \quad \forall i \in [m] \setminus Q, j \in Q; \quad (2)$$

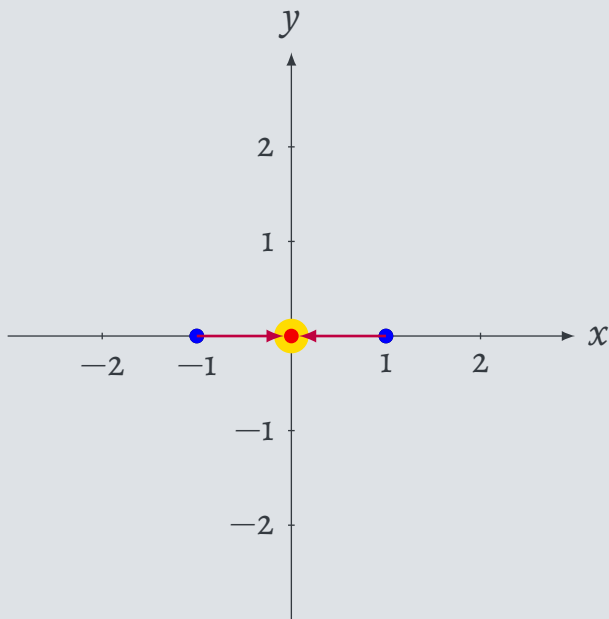
$$\alpha_i \geq \sum_{r=1}^m \lambda_r \sigma_{ir} d_{ij}, \quad \forall j \in Q, i \in \Gamma_j; \quad (3)$$

$$\alpha_i \leq \sum_{r=1}^m \lambda_r \sigma_{ir} d_{ij}, \quad \forall j \in Q, i \in [m] \setminus \Gamma_j. \quad (4)$$

Conversely, if σ verifies $d_{\sigma(r)} \geq d_{\sigma(r+1)}$, and $\exists \alpha \in \mathbb{R}^m$ verifying (1)-(4). Then, there exist $\tilde{u}, \tilde{v}$ such that $(\bar{y}, \bar{z}, \tilde{u}, \tilde{v})$ is optimal for DOMP_{IP}

If (2)-(4) are satisfied strictly, $\forall$ optimal solution (y', z', u', v') to DOMP_{LP} verifies $y' = \bar{y}$ and $z' = \bar{z}$

Degenerate Cases



But the optimal σ to DOMP_{LD} is

$$\sigma^* = \begin{pmatrix} 0 & 0 & 1 \\ 0.5 & 0.5 & 0 \\ 0.5 & 0.5 & 0 \end{pmatrix}$$

and every σ with full binary values is strictly suboptimal

Strongly Sortable Solutions

An optimal solution to DOMP_{LP} is said to be **strongly sortable solution** if the dual problem admits an optimal solution $(\alpha, \beta, \omega, \bar{\sigma})$ where $\bar{\sigma}$ takes binary values

Let $(\bar{y}, \bar{z})$ be a strongly sortable solution of DOMP_{LP} such that $(\alpha, \beta, \omega, \bar{\sigma})$ is an optimal solution to DOMP_{LD} where $\bar{\sigma} \in S_m$. Let $\sigma \in S_m$ any permutation fulfilling $d_{\sigma(r)} \geq d_{\sigma(r+1)}$ for $r \in [m-1]$. Then, there exists $\alpha' \in \mathbb{R}^m$ such that $(\alpha', \beta, \omega, \sigma)$ is an optimal solution to DOMP_{LD} . The set of points P is said to be **free-of-equidistance** if all entries above the main diagonal of its distance matrix are pairwise distinct

If P is free-of-equidistance, then, if $(\bar{y}, \bar{z}, u, v)$ is a mixed-integer optimal solution to DOMP_{LP} , the solution is strongly sortable

Main Result

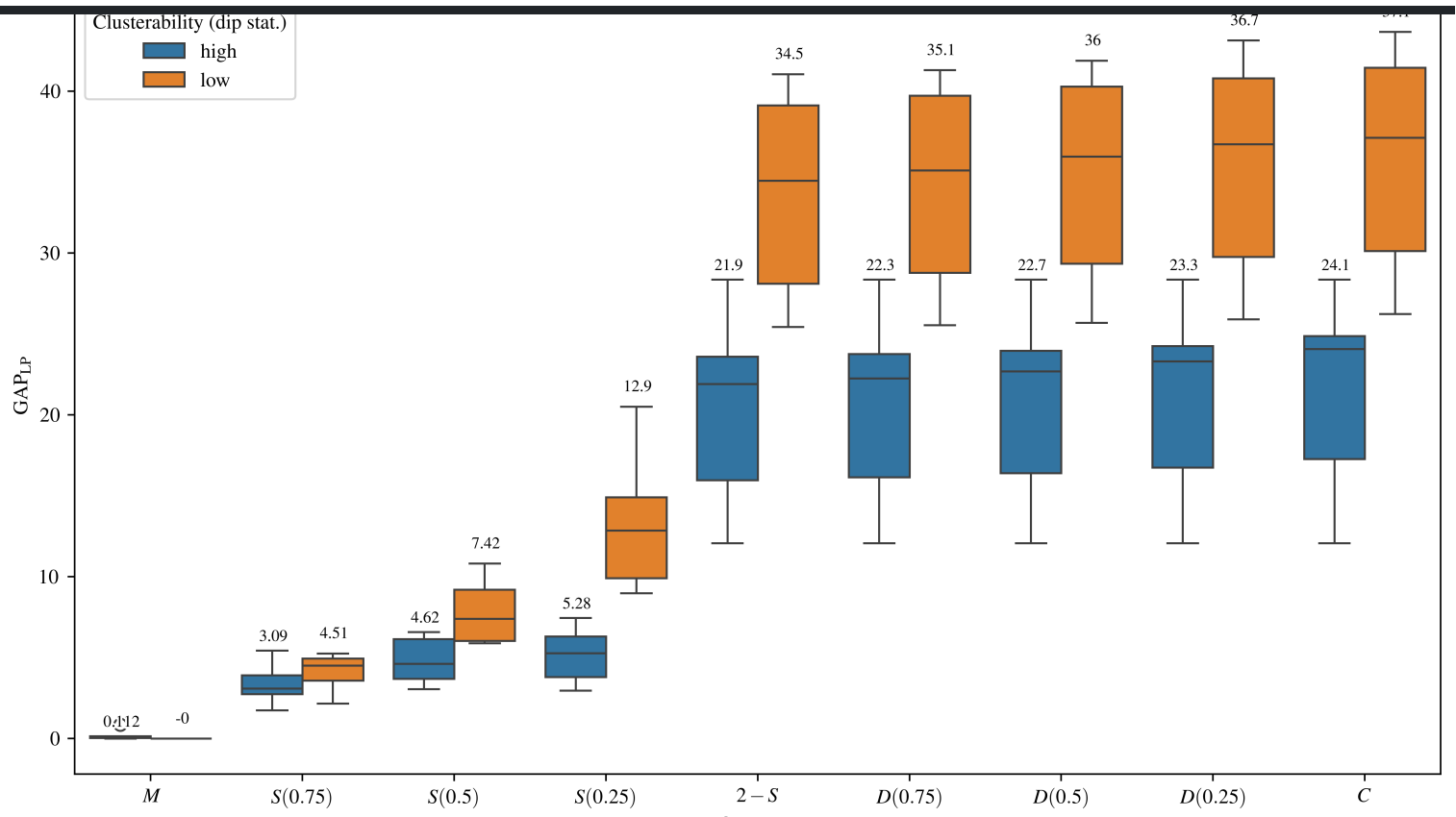
Theorem

Let $P \subset \mathbb{R}^n$ be a free-of-equidistance set of points and let p be a natural. Then, the following claims hold

1. If $\sum_{r=2}^m \lambda_r < \lambda_1$, the **p-DOMP** does not recover integrality
2. If no three points are collinear and $\sum_{r=2}^m \lambda_r = \lambda_1$, the **p-DOMP** does not recover integrality
3. M_1 recovers integrality (even if the points are non free of equidistance)
4. If no three points are collinear, S_p^2 does not recover integrality
5. C_p does not recover integrality
6. Let $0 \leq \gamma \leq 1$ be a real. If $\gamma(m-1) < 1$, the D_p^γ does not recover integrality

LP Relaxations and Clusterability

- p-med instances from ORLIB ($n \leq 400$).
- p -median, p -center, p - k -sums, and p - γ -centdians
- Empirical testing of non-recovery
- Impact of **clusterability** on the LP relaxations: **Hartigan and Hartigan (1985)** dip test classified instances on **low** and **highly** clusterable



INCORPORATING ORDERING INTO OPTIMIZATION PROBLEMS
○○○○

SCENARIO ANALYSIS IN LINEAR PROGRAMMING
○○○○○

WEIGHTED SET COVERING
○○○○○○○

TRAVELING SALESMAN PROBLEM
○○○○○○○

FITTING LINEAR MODELS
○○○○○○○○○

LP R
○○○

Conclusion

- We study, for the first time, the quality of LP relaxations in ordered location problems, deriving a novel primal–dual characterization that explains when the relaxation recovers integral solutions
- Through analytical results for several ordered models (p -median, p -center, p - γ -centdian, and p - k -sum), we show that the favorable behavior of the p -median LP relaxation does not extend to genuinely ordered problems
- Extensive computational experiments on ORLIB instances reveal that ordering significantly affects LP relaxation quality and that high data clusterability leads to relaxations closer to integer optima